Sahid Al Hassan

+91 8116282746 | sahidzack61@gmail.com | linkedin.com/in/sahid-al-hassan | github.com/me-sahid | zancrypt.in

EDUCATION

Chaitanya Deemed to be University — **Hyderabad, India**
Bachelor of Technology - Computer Science and Engineering; CGPA: 9.38 — Aug. 2024 – May 2028

Jakir Hossain Institute of Polytechnic — **West Bengal, India**
Diploma in Computer Science and Technology; CGPA: 8.00 — Aug. 2021 – May 2024

TECHNICAL SKILLS

Programming Languages: C++, C, Python, JavaScript, TypeScript, SQL

Databases & Caching: PostgreSQL, MySQL, Redis

Frameworks & Web: FastAPI, React.js, Next.js, Node.js, Tailwind CSS, HTML/CSS

Tools/Technologies: Docker, Linux, Git, Cloudflare, RESTful APIs

Core Concepts: Data Structures & Algorithms, Systems Engineering, Cybersecurity, Computer Networks

PROJECTS

Zancrypt [Link**]** | *Python, FastAPI, React, PostgreSQL, Redis, Docker* — May 12 – July 4, 2025

- Architected and deployed a production-grade **zero-knowledge cloud storage platform** featuring client-side **AES-256-GCM encryption** to ensure plain-text data never reaches the server.
- Engineered an asynchronous **FastAPI backend** utilizing SQLAlchemy, **Redis session management**, and secure HTTP-only refresh token cookies with JWT rotation.
- Designed distributed storage across S3 providers using **rendezvous hashing**, 3-way replication, and **SHA-256 integrity verification** with automatic rollbacks.
- Implemented passkey authentication, credential rotation, origin verification, and Razorpay payment integration.

Redis-Lite [Link**]** | *C, C++, Data Structures, Linux* — Aug. 5 – Aug. 23, 2026

- Engineered a high-performance, Redis-inspired **in-memory key-value database** in C/C++ supporting fundamental **SET, GET, and DELETE** operations.
- Designed dynamic **hash-table indexing** with custom collision handling and bucket management for low-latency memory lookup.
- Optimized raw pointer manipulations and memory allocations to eliminate memory leaks and ensure thread-safe key-value tracking.

AllIDo [Link**]** | *React, Node.js, PostgreSQL, Tailwind CSS* — 2026

- Developed a localized service marketplace connecting rural communities with verified local service providers to improve access in underserved regions.
- Designed end-to-end service discovery, booking workflows, and responsive UI layouts tailored specifically for low-bandwidth mobile networks.

CERTIFICATES / ACHIEVEMENTS

- **IEEE SSIT Hackathon:** Developed rapid technical prototypes under strict constraints in a competitive team setting at Vidya Jyothi Institute of Technology.
- **Hack2Skill Hackathon:** Collaborated on full-stack web application solutions addressing real-world technology challenges.
- **Academic Excellence:** Maintained a high academic standing with a **9.38 SGPA** at Chaitanya Deemed to be University.